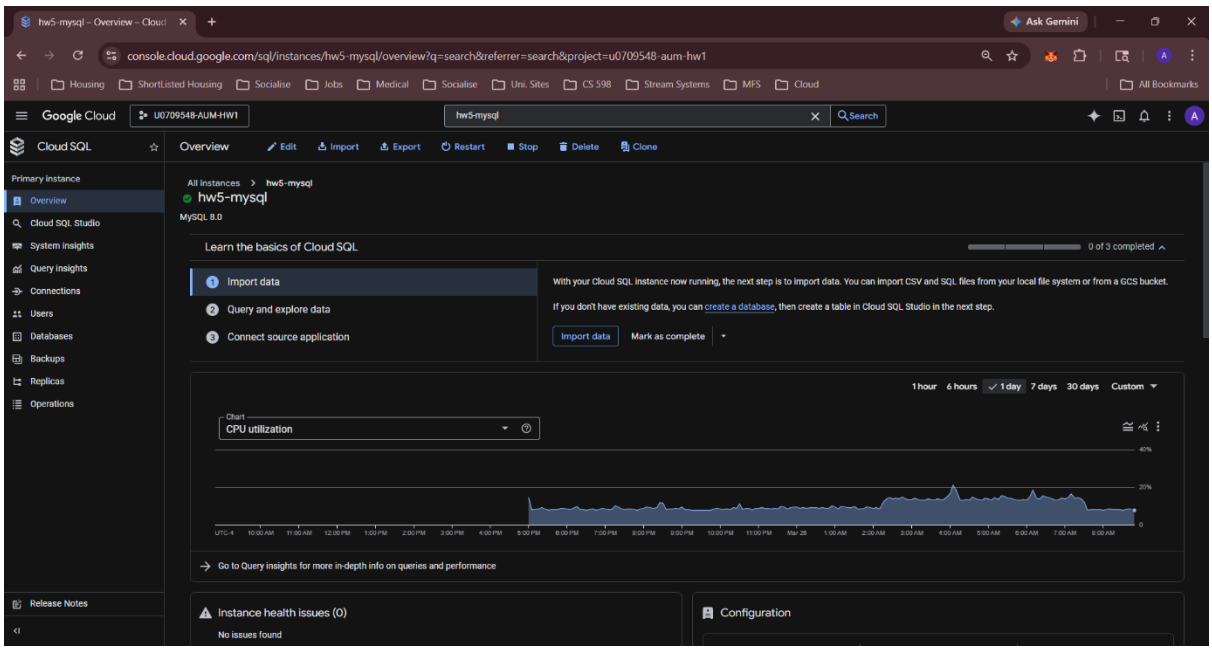
AUM – U07079548 – HW5 – CLOUD COMPUTING

Step 1:

HW5 builds on HW4's Service 1 and Service 2. Service 1 was modified to add Cloud SQL integration and timing instrumentation. Service 2 is unchanged.

Step 2 — Schema

Screenshot of Cloud SQL instance hw5-mysql in GCP Console (RUNNABLE)



Write the two tables:

requests: id, created_at, country, client_ip, gender, age, income, is_banned, time_of_day, requested_file, PRIMARY KEY(id)

errors: id, created_at, requested_file, error_code, PRIMARY KEY(id)

Step 3 —

Service 1 extracts X-country, X-gender, X-age, X-income, X-time, X-client-IP headers.

Successful requests insert into requests table. Failed requests (non-200) insert into errors table.

VM1 uses hw5-service1-sa with roles/cloudsql.client permission and Cloud SQL Auth Proxy v2 running as a systemd service.

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Step 4 — curl screenshots

```
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$ # 1. Successful request (200)
curl -v http://34.134.120.200/hw2/1.html

# 2. File not found (404)
curl -v http://34.134.120.200/hw2/filedoesnotexist.html

# 3. Forbidden country (403)
curl -v -H "X-country: Iran" http://34.134.120.200/hw2/1.html
*   Trying 34.134.120.200:80...
* Connected to 34.134.120.200 (34.134.120.200) port 80
* using HTTP/1.x
> GET /hw2/1.html HTTP/1.1
> Host: 34.134.120.200
> User-Agent: curl/8.10.1
> Accept: */*
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 26 Mar 2026 06:08:14 GMT
< Content-Type: text/html; charset=utf-8
< Content-Length: 117885
<
<!DOCTYPE html>
<html>
<body>
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit
 in voluptate velit esse
cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim
id est laborum.
<p>
<a HREF="2591.html"> This is a link </a>
<p>
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit
```

```
* shutting down connection #0
*   Trying 34.134.120.200:80...
* Connected to 34.134.120.200 (34.134.120.200) port 80
* using HTTP/1.x
> GET /hw2/filedoesnotexist.html HTTP/1.1
> Host: 34.134.120.200
> User-Agent: curl/8.10.1
> Accept: */*
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 404 Not Found
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 26 Mar 2026 06:08:16 GMT
< Content-Type: text/html; charset=utf-8
< Content-Length: 13
<
404 Not Found* shutting down connection #0
*   Trying 34.134.120.200:80...
* Connected to 34.134.120.200 (34.134.120.200) port 80
* using HTTP/1.x
> GET /hw2/1.html HTTP/1.1
> Host: 34.134.120.200
> User-Agent: curl/8.10.1
> Accept: */*
> X-country: Iran
>
* Request completely sent off
* HTTP 1.0, assume close after body
< HTTP/1.0 403 Forbidden
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Thu, 26 Mar 2026 06:08:17 GMT
< Content-Type: text/html; charset=utf-8
< Content-Length: 37
<
Forbidden: export restrictions apply.* shutting down connection #0
```

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HTTP/1.0 200 OK → curl http://34.134.120.200/hw2/1.html

HTTP/1.0 404 Not Found → curl http://34.134.120.200/hw2/filedoesnotexist.html

HTTP/1.0 403 Forbidden → curl -H "X-country: Iran" http://34.134.120.200/hw2/1.html

The DB output showing rows in both tables after the requests.

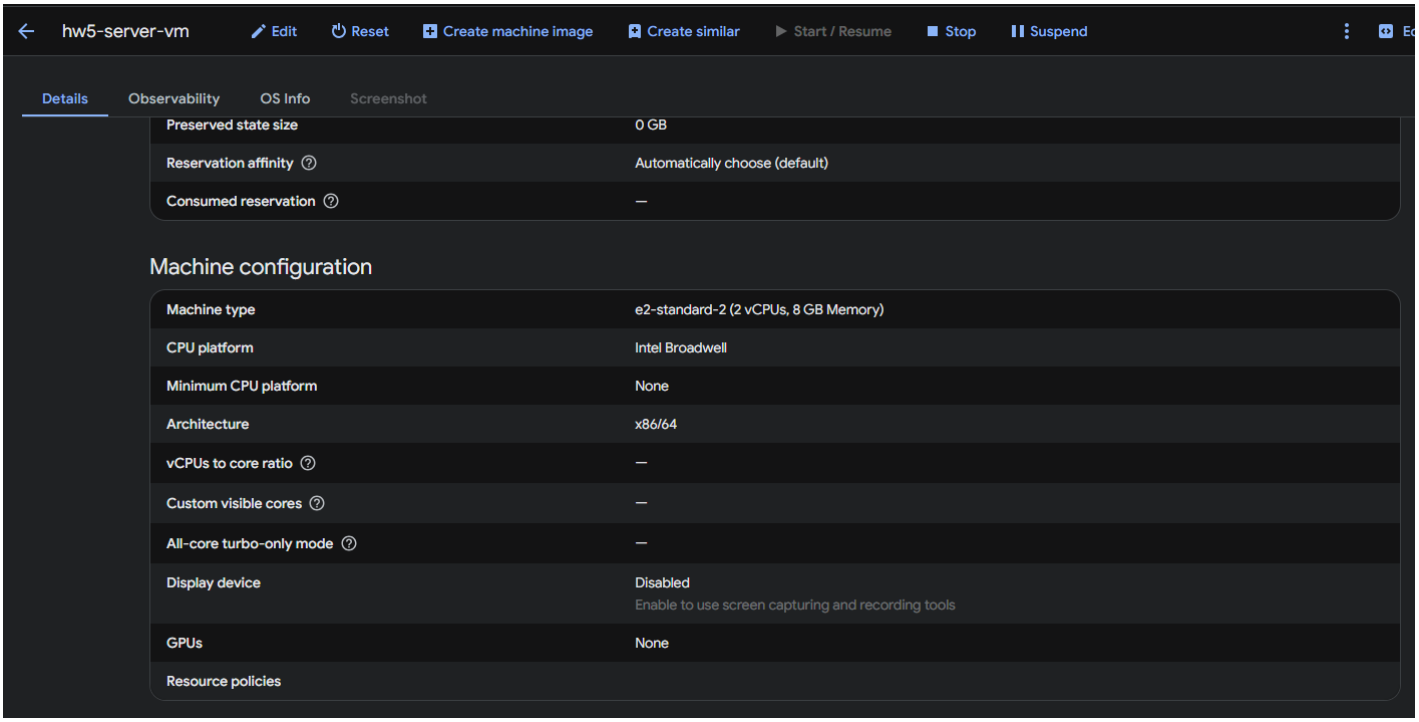
COMMAND:

```
gcloud compute ssh hw5-server-vm --zone=us-central1-a --project=u0709548-aum-hw1 --command="sudo -u www-data /opt/service1/venv/bin/python -c \"
import pymysql
c = pymysql.connect(host='127.0.0.1',port=3306,user='hw5user',password='aumcloudhw123',database='hw5db',cursorclass=pymysql.cursors.DictCursor)
cur = c.cursor()
cur.execute('SELECT id,country,is_banned,requested_file,created_at FROM requests ORDER BY id DESC LIMIT 5')
print('=== requests table ===')
for r in cur.fetchall(): print(r)
cur.execute('SELECT id,requested_file,error_code,created_at FROM errors ORDER BY id DESC LIMIT 5')
print('=== errors table ===')
for r in cur.fetchall(): print(r)
c.close()
\"" 2>&1
```

```
=== requests table ===
{'id': 4, 'country': 'Iran', 'is_banned': 1, 'requested_file': '/hw2/1.html', 'created_at': datetime.datetime(2026, 3, 26, 6, 8, 17, 581000)}
{'id': 3, 'country': '', 'is_banned': 0, 'requested_file': '/hw2/1.html', 'created_at': datetime.datetime(2026, 3, 26, 6, 8, 14, 645000)}
{'id': 2, 'country': '', 'is_banned': 0, 'requested_file': '/hw2/1.html', 'created_at': datetime.datetime(2026, 3, 26, 6, 7, 19, 238000)}
{'id': 1, 'country': '', 'is_banned': 0, 'requested_file': '/hw2/1.html', 'created_at': datetime.datetime(2026, 3, 26, 6, 7, 4, 880000)}
=== errors table ===
{'id': 15, 'requested_file': '/hw2/1.html', 'error_code': 403, 'created_at': datetime.datetime(2026, 3, 26, 6, 8, 17, 587000)}
{'id': 14, 'requested_file': '/hw2/filedoesnotexist.html', 'error_code': 404, 'created_at': datetime.datetime(2026, 3, 26, 6, 8, 16, 309000)}
{'id': 13, 'requested_file': '/hw2/1.html', 'error_code': 500, 'created_at': datetime.datetime(2026, 3, 26, 6, 4, 56, 590000)}
{'id': 12, 'requested_file': '/hw2/1.html', 'error_code': 500, 'created_at': datetime.datetime(2026, 3, 26, 6, 3, 40, 839000)}
{'id': 11, 'requested_file': '/hw2/1.html', 'error_code': 500, 'created_at': datetime.datetime(2026, 3, 26, 6, 0, 59, 701000)}
```

Step 5 — VM + 2 clients

Screenshot of VM1 in GCP Console showing e2-standard-2



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Screenshot of both client terminals running simultaneously

```
MINGW64:/d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COM...

Aum@DESKTOP-1KD3K3T MINGW64 ~
$ ./http-client.exe -d 34.134.120.200 -b aum-hw2-u07079548 -w hw2 -n 50000 -i 20
000 -p 80 -r 42
bash: ./http-client.exe: No such file or directory

Aum@DESKTOP-1KD3K3T MINGW64 ~
$ cd "/d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-C
LOUD-COMPUTING-HW/hw5"

Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Githu
b Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$ ./http-client.exe -d 34.134.120.200 -b aum-hw2-u07079548 -w hw2 -n 50000 -i 20
000 -p 80 -r 42

Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Githu
b Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$
```

```
MINGW64:/d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTI...

Aum@DESKTOP-1KD3K3T MINGW64 ~
$ ./http-client.exe -d 34.134.120.200 -b aum-hw2-u07079548 -w hw2 -n 50000 -i 20
000 -p 80 -r 42
bash: ./http-client.exe: No such file or directory

Aum@DESKTOP-1KD3K3T MINGW64 ~
$ cd "/d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-C
LOUD-COMPUTING-HW/hw5"

Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Githu
b Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$ ./http-client.exe -d 34.134.120.200 -b aum-hw2-u07079548 -w hw2 -n 50000 -i 20
000 -p 80 -r 42

Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Githu
b Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$
```

Write: "VM1 uses e2-standard-2 (2 vCPUs, 8GB RAM) to handle 2 concurrent clients. Both clients used seed 42 and sent 50,000 requests each (100,000 total)."

Step 6 — Timing instrumentation

Service 1 instruments 4 separate operations using time.perf_counter_ns():

extract_headers() — time to parse HTTP request headers

read_file_from_gcs() — time to download file from Cloud Storage

send_response_body() — time to write HTTP response to client

insert_request() / insert_error() — time to insert row into Cloud SQL

Timings are logged to Cloud Logging as REQUEST_TIMING events in milliseconds.

```
2026-03-26 08:56:41.295  hw5-server-vm  {"db_insert_ms":6.6919, "event":"REQUEST_TIMING", "gcs_read_ms":0, "headers_ms":0.006165, "send_response_ms":0.189802}
Investigate log   Copy   Expand nested fields   Hide log summary
{
  insertId: "1acr0hjf7uh2j3"
  jsonPayload: {
    db_insert_ms: 6.6919
    event: "REQUEST_TIMING"
    gcs_read_ms: 0
    headers_ms: 0.006165
    send_response_ms: 0.189802
  }
  logName: "projects/u0709548-aum-hw1/logs/hw5-service1"
  receiveTimestamp: "2026-03-26T12:56:41.295937774Z"
  resource: {2}
  severity: "INFO"
  timestamp: "2026-03-26T12:56:41.295937774Z"
}
```

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Step 7 — Statistics

COMMAND : DB_HOST=136.113.97.153 DB_NAME=hw5db DB_USER=hw5user DB_PASSWORD=aumcloudhw123 /c/Python313/python.exe query_stats.py

```
Connected to 136.113.97.153:3306/hw5db

============================================================
1. Successful vs Unsuccessful requests:
   Successful   : 95402
   Unsuccessful : 4768

2. Requests from banned countries: 4768

3. Requests by gender:
   Female      : 50461
   Male        : 49705
   (unknown)   : 4

4. Top 5 countries sending requests:
   1. Albania                      : 589
   2. Burkina Faso                 : 587
   3. Greece                       : 576
   4. Myanmar                      : 576
   5. Iran                         : 570

5. Age group with most requests: 26-35 (12804 requests)

6. Income group with most requests: 10k-20k (12734 requests)
```

Successful: 95,402 / Unsuccessful: 4,768

Requests from banned countries: 4,768

Male: 49,705 / Female: 50,461

Top 5 countries: Albania, Burkina Faso, Greece, Myanmar, Iran

Age group with most requests: 26-35 (12,804 requests)

Income group with most requests: 10k-20k (12,734 requests)

Step 8 — Cloud Function (Could not capture SS from GCP since I had already ran the cleanup script)

Screenshot of stop-idle-db Cloud Function in GCP Console (ACTIVE) from the setup script

Screenshot of Cloud Scheduler job showing 0 * * * * from the setup script

```
state: ACTIVE
updateTime: '2026-03-26T05:28:04.400032969Z'
url: https://us-central1-u0709548-aum-hw1.cloudfunctions.net/stop-idle-db
attemptDeadline: 180s
httpTarget:
  headers:
    User-Agent: Google-Cloud-Scheduler
  httpMethod: POST
  oidcToken:
    audience: https://stop-idle-db-zgn3xxujtq-uc.a.run.app/
    serviceAccountEmail: hw5-cf-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  uri: https://stop-idle-db-zgn3xxujtq-uc.a.run.app/
name: projects/u0709548-aum-hw1/locations/us-central1/jobs/stop-idle-db-scheduler
retryConfig:
  maxBackoffDuration: 3600s
  maxDoublings: 5
  maxRetryDuration: 0s
  minBackoffDuration: 5s
schedule: 0 * * * *
scheduleTime: '2026-03-26T06:00:00Z'
state: ENABLED
status:
  code: -1
timeZone: Etc/UTC
userUpdateTime: '2026-03-26T05:28:15.863419Z'
```

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A Cloud Function checks hourly if the DB is RUNNABLE and stops it by setting activationPolicy=NEVER. Triggered by Cloud Scheduler every hour to prevent unnecessary billing.

Step 9 — Setup/Cleanup scripts

Run bash setup.sh from the hw5/ directory. Project ID is hardcoded as u0709548-aum-hw1. The script creates Cloud SQL, Pub/Sub, service accounts, VMs, and deploys the Cloud Function automatically. Run bash cleanup.sh to tear down all resources.

Setup Script Run:

```
Aum@DESKTOP-1KD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$ bash setup.sh
=== HW5 Setup ===
Project : u0709548-aum-hw1 (1097067927322)
Region  : us-central1 / us-central1-a
[1/10] Enabling APIs ...
Operation "operations/acat.p2-1097067927322-e03babc8-225c-4cd2-a3e0-a6451f191dd5" finished successfully.
[2/10] Uploading scripts to GCS ...
Copying file://service1.py to gs://aum-hw2-u07079548/hw5-scripts/service1.py
  Completed files 1/1 | 13.8kiB/13.8kiB
Copying file://service2.py to gs://aum-hw2-u07079548/hw5-scripts/service2.py
  Completed files 1/1 | 1.7kiB/1.7kiB
Copying file://setup_schema.py to gs://aum-hw2-u07079548/hw5-scripts/setup_schema.py
  Completed files 1/1 | 3.6kiB/3.6kiB
[3/10] Configuring Cloud SQL instance 'hw5-mysql' ...
  Instance state: RUNNABLE — nothing to do.
  DB_CONN_NAME=u0709548-aum-hw1:us-central1:hw5-mysql
  DB_PUBLIC_IP=136.113.97.153
  Authorizing setup machine IP 73.61.150.179 for Cloud SQL ...
The following message will be used for the patch API method.
{"name": "hw5-mysql", "project": "u0709548-aum-hw1", "settings": {"ipConfiguration": {"authorizedNetworks": [{"value": "73.61.150.179/3
2"}]}}}
Patching Cloud SQL instance...done.
Updated [https://sqladmin.googleapis.com/sql/v1beta4/projects/u0709548-aum-hw1/instances/hw5-mysql].
  Running schema setup from setup.sh ...
  Using Python: /c/Python313/python.exe

[notice] A new release of pip is available: 25.0.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
[schema] Connecting to MySQL at 136.113.97.153:3306 ...
[schema] Database 'hw5db' ready.
[schema] Table 'requests' created (or already exists).
[schema] Table 'errors' created (or already exists).
[schema] Schema setup complete.
[4/10] Creating Pub/Sub topic and subscription ...
  Topic already exists.
  Subscription already exists.
```

```
[4/10] Creating Pub/Sub topic and subscription ...
  Topic already exists.
  Subscription already exists.
[5/10] Creating service accounts and IAM bindings ...
  SA hw5-service1-sa already exists.
Updated IAM policy for project [u0709548-aum-hw1].
bindings:
- members:
  - serviceAccount:service-1097067927322@gcp-sa-artifactregistry.iam.gserviceaccount.com
  role: roles/artifactregistry.serviceAgent
- members:
  - serviceAccount:1097067927322@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcf-admin-robot.iam.gserviceaccount.com
  role: roles/cloudfunctions.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudscheduler.iam.gserviceaccount.com
  role: roles/cloudscheduler.serviceAgent
- members:
  - serviceAccount:hw5-service1-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/cloudsql.client
- members:
  - serviceAccount:1097067927322@cloudservices.gserviceaccount.com
  role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-1097067927322@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:service-1097067927322@containerregistry.iam.gserviceaccount.com
  role: roles/containerregistry.ServiceAgent
- members:
  - serviceAccount:1097067927322-compute@developer.gserviceaccount.com
  - serviceAccount:1097067927322@cloudservices.gserviceaccount.com
  - serviceAccount:u0709548-aum-hw1@appspot.gserviceaccount.com
```

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```
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  - serviceAccount:hw5-service1-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  - serviceAccount:hw5-service2-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/storage.objectViewer
etag: BwZN5p_X8cY=
version: 1
[6/11] Reserving static IP ...
  Static IP: 34.134.120.200
[7/11] Creating firewall rule ...
NAME            NETWORK  DIRECTION  PRIORITY  ALLOW   DENY  DISABLED
hw5-allow-http  default  INGRESS    1000      tcp:80        False
[8/10] Creating VM1 (web server) ...
Created [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw5-server-vm].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce the risk of wide
spread service disruption, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME           ZONE           MACHINE_TYPE   PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP     STATUS
hw5-server-vm  us-central1-a  e2-standard-2               10.128.0.7   34.134.120.200  RUNNING
  VM1 (hw5-server-vm) created at 34.134.120.200:80
[9/10] Creating VM3 (service 2 reporter) ...
Created [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw5-service2-vm].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce the risk of wide
spread service disruption, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME            ZONE           MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP  STATUS
hw5-service2-vm  us-central1-a  e2-micro                  10.128.0.8   34.46.23.31  RUNNING
[10/10] Deploying Cloud Function 'stop_idle_db' ...
Preparing function...done.
OK Deploying function...
  OK [Build] Logs are available at [https://console.cloud.google.com/cloud-build/builds;region=us-central1/51150f24-0813-4963-942c-70a
  489b4dab6?project=1097067927322]
  OK [Service]
  .  [ArtifactRegistry]
  .  [Healthcheck]
  .  [Triggercheck]
Done.
```

```
url: https://us-central1-u0709548-aum-hw1.cloudfunctions.net/stop-idle-db
attemptDeadline: 180s
httpTarget:
  headers:
    User-Agent: Google-Cloud-Scheduler
  httpMethod: POST
  oidcToken:
    audience: https://stop-idle-db-zgn3xxujtq-uc.a.run.app/
    serviceAccountEmail: hw5-cf-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  uri: https://stop-idle-db-zgn3xxujtq-uc.a.run.app/
name: projects/u0709548-aum-hw1/locations/us-central1/jobs/stop-idle-db-scheduler
retryConfig:
  maxBackoffDuration: 3600s
  maxDoublings: 5
  maxRetryDuration: 0s
  minBackoffDuration: 5s
schedule: 0 * * * *
scheduleTime: '2026-03-26T06:00:00Z'
state: ENABLED
status:
  code: -1
timeZone: Etc/UTC
userUpdateTime: '2026-03-26T05:28:15.863419Z'

=== HW5 Setup Complete ===
  Server (VM1) static IP   : 34.134.120.200
  Server port              : 80
  Cloud SQL instance       : hw5-mysql (u0709548-aum-hw1:us-central1:hw5-mysql)

  NOTE: VM1 may take 2-3 minutes to finish installing dependencies.
  Test with:
    curl http://34.134.120.200/hw2/<filename>

  To run the 2 concurrent clients (from your local Windows laptop):
  Open 2 terminals and run the same command in both simultaneously:
    ./http-client.exe -d 34.134.120.200 -b aum-hw2-u07079548 -w hw2 -n 50000 -i 20000 -p 80 -r 42

  After clients finish, run stats:
    DB_HOST=136.113.97.153 DB_NAME=hw5db DB_USER=hw5user DB_PASSWORD=aumcloudhw@123 python3 query_stats.py
```

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Cleanup Script:

```
Aum@DESKTOP-IKD3K3T MINGW64 /d/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw5 (main)
$  bash cleanup.sh
=== HW5 Cleanup ===
Project: u0709548-aum-hw1
[1/9] Deleting VM instances ...
[2/9] Releasing static IP ...
[3/9] Deleting firewall rule ...
[4/9] Deleting Pub/Sub resources ...
[5/9] Deleting Cloud Scheduler job ...
[6/9] Deleting Cloud Function ...
[7/9] Stopping Cloud SQL instance 'hw5-mysql' (NOT deleting) ...
The following message will be used for the patch API method.
{"name": "hw5-mysql", "project": "u0709548-aum-hw1", "settings": {"activationPolicy": "NEVER"}}
Patching Cloud SQL instance...done.
Updated [https://sqladmin.googleapis.com/sql/v1beta4/projects/u0709548-aum-hw1/instances/hw5-mysql].
  Cloud SQL instance stopped.
[8/9] Removing IAM bindings and service accounts ...
bindings:
- members:
  - serviceAccount:service-1097067927322@gcp-sa-artifactregistry.iam.gserviceaccount.com
  role: roles/artifactregistry.serviceAgent
- members:
  - serviceAccount:1097067927322@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcf-admin-robot.iam.gserviceaccount.com
  role: roles/cloudfunctions.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudscheduler.iam.gserviceaccount.com
  role: roles/cloudscheduler.serviceAgent
- members:
  - serviceAccount:hw5-cf-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/cloudsql.admin
- members:
  - serviceAccount:hw5-service1-sa@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/cloudsql.client
```

```
- members:
  - serviceAccount:service-1097067927322@containerregistry.iam.gserviceaccount.com
  role: roles/containerregistry.ServiceAgent
- members:
  - serviceAccount:1097067927322-compute@developer.gserviceaccount.com
  - serviceAccount:1097067927322@cloudservices.gserviceaccount.com
  - serviceAccount:u0709548-aum-hw1@appspot.gserviceaccount.com
  role: roles/editor
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/logging.logWriter
- members:
  - user:aum1612@bu.edu
  role: roles/owner
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/pubsub.publisher
- members:
  - serviceAccount:service-1097067927322@gcp-sa-pubsub.iam.gserviceaccount.com
  role: roles/pubsub.serviceAgent
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/pubsub.subscriber
- members:
  - serviceAccount:service-1097067927322@serverless-robot-prod.iam.gserviceaccount.com
  role: roles/run.serviceAgent
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/storage.objectViewer
etag: BwZN7a5bS-Y=
version: 1
[9/9] Removing hw5-scripts from GCS ...

=== HW5 Cleanup Complete ===
  Cloud SQL 'hw5-mysql' has been STOPPED (not deleted).
  HW2 bucket data files were NOT deleted.
```

Github Link - https://github.com/aumghelani/AUM-U07079548-CLOUD-COMPUTING-HW/tree/main/hw5

https://github.com/aumghelani/AUM-U07079548-CLOUD-COMPUTING-HW.git